

INFLUENCE OF MUSCLE–TENDON UNIT STRUCTURE ON RATE OF FORCE DEVELOPMENT DURING THE SQUAT, COUNTERMOVEMENT, AND DROP JUMPS

JACOB E. EARP,¹ WILLIAM J. KRAEMER,² PRUE CORMIE,¹ JEFFERY S. VOLEK,² CARL M. MARESH,² MICHAEL JOSEPH,² AND ROBERT U. NEWTON¹

¹*School of Exercise, Biomedical and Health Sciences, Edith Cowan University, Joondalup, WA, Australia; and* ²*Human Performance Laboratory, Department of Kinesiology, University of Connecticut, Storrs, Connecticut*

ABSTRACT

Earp, JE, Kraemer, WJ, Cormie, P, Volek, JS, Maresch, CM, Joseph, M, and Newton, RU. Influence of muscle–tendon unit structure on rate of force development during the squat, countermovement, and drop jumps. *J Strength Cond Res* 25(2): 340–347, 2011—Previous research has highlighted the importance of muscle and tendon structure to stretch shortening cycle performance. However, the relationships between muscle and tendon structure to performance are highly dependent on the speed and intensity of the movement. The purpose of this study was to determine if muscle and tendon structure is associated with the rate of force development (RFD) throughout static squat jump (SJ), countermovement jump (CMJ), and drop jump (DJ; 30-cm height). Twenty-five strength- and power-trained men participated in the study. Using ultrasonography, vastus lateralis (VL) and gastrocnemius (GAS) pennation (PEN) and fascicle length (FL), and Achilles tendon (AT) thickness and length were measured. Subjects then performed SJ, CMJ, and DJ, during which RFD was calculated over time 5 distinct time intervals. During CMJs, early RFD could be predicted between 0 and 10 milliseconds by both GAS-FL ($r^2 = 0.213$, $\beta = 0.461$) and AT-length ($r^2 = 0.191$, $\beta = -0.438$). Between 10 and 30 milliseconds GAS-FL was a significant predictor of CMJ-RFD ($r^2 = 0.218$, $\beta = 0.476$). During DJ, initial RFD (0–10 milliseconds) could be significantly predicted by GAS-FL ($r^2 = 0.185$, $\beta = -0.434$), VL-PEN ($r^2 = 0.189$, $\beta = 0.435$), and GAS-PEN ($r^2 = 0.188$, $\beta = 0.434$). These findings suggest that longer ATs may have increased elasticity, which can decrease initial RFD during CMJ; thus, their use in talent identification is not recommended. The GAS fascicle length

had an intensity-dependent relationship with RFD, serving to positively predict RFD during early CMJs and an inverse predictor during early DJs. During DDJs, subjects with greater PEN were better able to redirected initial impact forces. Although both strength and plyometric training have been shown to increase FL, only heavy strength training has been shown to increase PEN. Thus, when a high eccentric load or multiple jumps are required, heavy strength training might be used to elicit muscular adaptations that are suited to fast force production during jumping.

KEY WORDS vertical jump, fascicle, pennation, achilles tendon, tendon length, impulse

INTRODUCTION

During stretch shortening cycle (SSC) movements, such as vertical jumping, the ability to produce force rapidly is vital to athletic performance. Understanding the relationship between muscle and tendon structure and these abilities can increase the efficacy of talent identification and provide practitioners with the ability to prescribe specific strength and conditioning programs that focus on specific structural adaptations rather than simply mimicking movement patterns and velocities.

Muscle structure is very important to the expression of force during movement. Muscle thickness has been found to be highly related to an individual's ability to produce force (22). The architecture of the muscle becomes increasingly important as the speed of movement changes (4,5). Muscle pennation (PEN) is the angle at which muscle fascicles are oriented relative to the tendon to which they attach (Figure 1) and is believed to be able to increase force production capabilities during low-velocity movements by increasing the physiological cross-sectional area of a muscle (4,5,14,23,26). In contrast, greater muscle fascicle length (FL) results in more contractile elements in series, which increases the ability to produce force during high velocity contractions and allows for less sarcomere deformation at a given velocity (5,8). Muscle architecture has been shown to be highly

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Address correspondence to Jacob E. Earp, j.earp@ecu.edu.au.
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adaptable to resistance training (1,3,6,7,9,10). High force, low-velocity training has been shown to result in increases in muscle cross-sectional thickness, angle of PEN, and muscle FL (1,3,9). In contrast lower force, high-velocity training is more often associated with increases in muscle FL, decreases in PEN, and little or no changes in muscle thickness (2,10).

Using ultrasonography FL and PEN adaptation to resistance training have become detectable in as little as 5 weeks, even before any detectable changes in muscle thickness (6,9). Blazevich et al. (6) in a 10-week training study with untrained subjects found that vastus lateralis (VL) FL quickly increased in the first 5 weeks of training and then reached a plateau, whereas PEN increased incrementally throughout the entire 10 weeks. Of the studies mentioned, only one used an altered resistance training stimuli with trained athletes (9). In this study, athletes who performed either resistance training and squat jump/sprint training or only squat jump/sprint training increased VL thickness. However VL-FL increased only with the squat jump/sprint training group and VL-PEN increased in the resistance training compared to the squat jump/sprint group. These differences in training-specific structural adaptations are important to understand when designing a strength and conditioning training program so that an optimal stimuli can be used to elicit the structural adaptations most beneficial to athletic performance.

To our knowledge, only 1 study has investigated the role of muscle architecture on RFD (7). In this study, Blazevich et al. (7) found that after 5 weeks of heavy resistance training with untrained subjects there was an inverse relationship between changes in FL and early RFD (0–30 milliseconds) during an isometric knee extension. This inverse relationship was attributed to increased elastic properties of the muscle, despite the increase in potential contractile velocity that resulted from the increases in FL. Currently, there is a need for research to examine the relationship between muscle architecture and RFD during SSC movements such as vertical jumping.

Muscle architecture has been found to be highly associated with SSC performance in movements such as sprinting and jumping performance (2,15,25). Previous research has observed that competitive sprinters with >100-m running times possessed longer FL and lower PEN in their gastrocnemius (GAS) and VL muscles (25). However, caution is to be exercised when comparing sprinting and jumping movements because SSC duration and the action of the GAS differ between these movements. In a vertical jumping, Earp et al. (15) reported that greater GAS-PEN was a significant predictor of jump performance during SJ, countermovement jump (CMJ) and drop jump (DJ). In addition, it was also observed that subjects with long GAS-FL had decreased jump performance during the DJ when compared to the CMJ and SJ. This performance detriment was attributed to an uncontrolled elongation of the plantar flexor muscle (15). Such uncontrolled elongation has previously been reported to occur in individuals who

performed jumps from drop heights that lead to impact forces greater than which the plantar flexors could tolerate (20,31).

Achilles tendon (AT) length has long been used as a marker for talent identification for jumping athletes, with many coaches anecdotally believing that athletes with long tendons will be better equipped for performance. However, such beliefs are currently unsubstantiated. A modeling study by Nagano et al. (28) found that long tendons should be better suited for projecting a given load while the mass being projected is low. But, when the mass being projected increases, shorter tendons were better suited for performance because they shifted from working as a power amplifier to a force transducer (28). Tendon structure is highly related to tendon compliance in that long thin tendons are more compliant than short thick tendons (11,12,30,32). During isometric contractions, RFD has been shown to decrease with increased AT length (32), however, as yet no such relationships have been found with complex SSC movements such as vertical jumping.

During vertical jumping, the relationship between tendon structure and jump performance is complex. Bobbert (11) found that subjects with long compliant ATs performed better during static squat jumping (SJ) than did subjects with short stiff ATs. Bobbert (11) attributed this to increased time to reach maximal muscle tension because of the greater tendon deformation resulting from initial force expression. However, if such is the case, it is possible that RFD may be decreased via the same process which could negatively affect athletic performance.

Although jump and sprint running performances are important factors influencing sporting success, it is important to note that it is the ability to produce force quickly throughout a variety of SSC tasks that often determines the outcome of play. A better understanding of the anatomical predictors of these movements can result in a more accurate talent identification and improved training program design. Thus, it is the purpose of this study to determine whether muscle and tendon structure can predict force expression during vertical jumping with 3 different prejump loads (SJ, CMJ, and DJ). It is hypothesized that the muscle and tendon structure most beneficial to performance will vary based on the time phase of the jump and the prejump load.

METHODS

Experimental Approach to the Problem

This study used a cross-sectional experimental design in which 25 trained men were assessed for lower-body muscle-tendon unit structure and force expression throughout 3 types of jumps each with different prejump loading. Muscle structure of the VL and lateral GAS and tendon structure of the patellar tendon (PT) and AT were determined using ultrasonography. Then RFD was calculated between 6 different time points during SJ, CMJ, and 30-cm DJ. Lastly,

statistical analyses were used to determine if muscle and tendon structure could predict how force was produced throughout jumping and how intensity of the countermovement influences these relationships.

Before all testing, subjects were required to refrain from any intense or unaccustomed exercise for at least 48 hours and any exercise at all for at least 24 hours. Additionally, subjects abstained from caffeine and alcohol for at least 12 hours before testing and encouraged to standardize their diet and be euhydrated during testing. A separate analysis of the same data looking at jump performance has previously been published (15).

Subjects

Twenty-five strength and power-trained men, between the ages of 18 and 35, volunteered for the study (Table 1). All subjects had been consistently strength training, at least 2 times per week, for the last 6 months and also trained regularly with speed, plyometric, and power training. Subjects came from a variety of sporting and training backgrounds (i.e., Weightlifting, Power Lifting, American football, track and field, volley, and recreational training). This population was selected to represent a cross-section of athletic populations with a variety of body types. Before participation in the study, each subject was familiarized with testing procedures and allowed to practice with each test until they felt comfortable with the movements and used proper form as deemed by the researchers; no feedback to form was given during testing. This experiment was approved by the University of Connecticut’s institutional review board, and all subjects were informed of the procedures and potential risks involved before they gave their written consent to participate.

Muscle–Tendon Structure

Muscle–tendon unit structure was assessed using ultrasonography (16 Hz, 3.5 MHz, Aspen Advanced: ACUSON, Monsey, NY, USA). Previous research has demonstrated these measurements to be both valid and reliable (22). All measurements were taken on the subject’s self-reported dominant side. Three separate longitudinal images were taken, and 3 measurements were taken from each image and the average reported. Muscle thickness, PEN, and FL were assessed for VL and GAS (Figure 1). The VL measurements were taken at one-half the length from the greater trochanter to the lateral joint space of the knee midhigh length with the subjects lying supine with their leg fully extended and relaxed. The

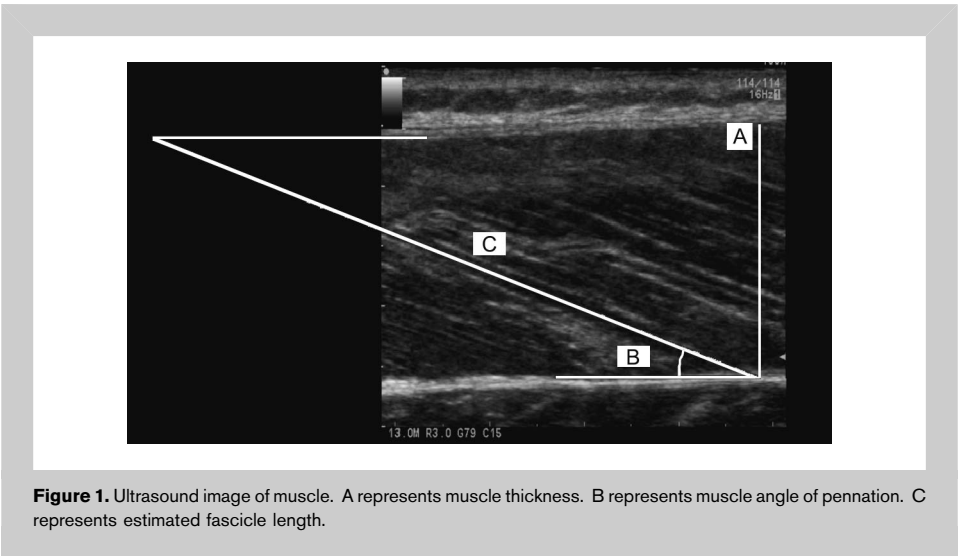
TABLE 1. Characteristics of the experimental subjects (n = 25).

Age	Height (cm)	Weight (kg)	Training age (y)
23.3 ± 3.2	176.1 ± 7.4	86.2 ± 11.6	5.7 ± 4.7

GAS measurements were taken at two-thirds the lower leg length from the lateral malleolus of the tibia to the lateral joint space of the knee with the subjects lying prone with their foot unloaded and unsupported. When an entire muscle fascicle could not be viewed in the same image, FL was calculated using the equation below (16).

Fascicle Length = Muscle Thickness
× Cosine (Pennation Angle).

To assess tendon thickness of the PT, subjects would lay supine with their knee and hip flexed to 90°. Each image was taken so that the inferior border of the patella and the transverse tibial tuberosity were visible in the same image and PT thickness was calculated at half the distance between these 2 anatomical landmarks (Figure 2A). Achilles Tendon measurements were taken with the subjects lying prone with their foot supported. Because of the taper of the AT as it rises superiorly, AT thickness was assessed at 2.5 cm above the inferior dorsal border of the calcaneus. To determine AT length, ultrasonography was used to trace the AT superiorly starting from the inferior dorsal border of the calcaneus toward the muscle belly of the GAS (Figure 2B). The position of the myotendinous junction was determined using



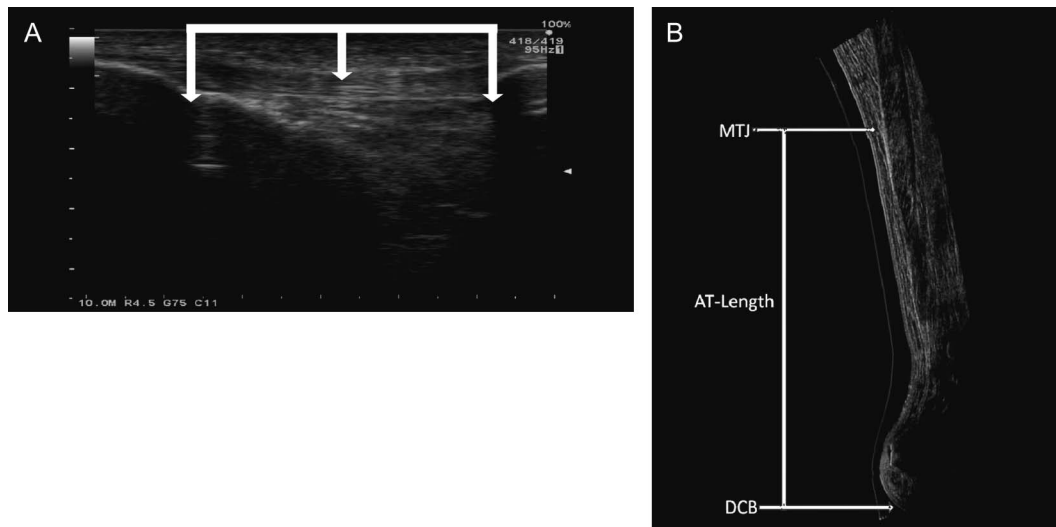


Figure 2. Ultrasound images displaying the location of measurements of Patellar Tendon Thickness (A) and Achilles Tendon Length (B). MTJ = Myotendinous Junction; AT = Achilles Tendon; DCB = Distal Calcaneal Boarder.

ultrasonography; this position was then marked on the skin and the displacement of the probe was used to determine AT length.

Vertical Jump Testing

Before vertical jump testing, subjects performed a standardized warm-up consisting of low-intensity cycle ergometry and dynamic lower-body stretches before vertical jump testing. The dynamic stretches included body weight squats, knee hugs, walking lunges, walking quadriceps stretch, high kicks, and lateral lunges. Subjects received verbal encouragement during testing, which remained consistent between all subjects and trials.

Vertical jump performance was analyzed for 3 types of jump with different prejump loads: SJ (low), CMJ (moderate), and DJ (high). Subjects performed 2 sets of 3 repetitions for each jump type in a random and counterbalanced order. Each set was separated by 3 minutes to allow for full recovery, and no order or fatigue affect was found between sets. All jumps were performed with hands on hips to limit use of upper body.

Static squat jumps were defined as jumps in which the subjects would lower themselves to the bottom position of their jump, pause for 2 seconds, and then jumped as high as possible with no prior countermovement. The bottom position of SJs was based on the self-selected position observed during previous CMJs and practiced several times before the first set of SJ. Jumps that exhibited any countermovement in the force data were not included and a replacement jump required. Countermovement jumps were defined as those jumps in which the subject started in a standing position then dropped down to a volitional depth and jumped as high as possible. Drop jumps were defined as

jumps in which the subjects would step off a 30-cm box with 1 foot, land on both feet, and jump as high as possible.

Video analysis was used to determine vertical jump height and joint angles for each jump (DartFish ProSuit 5.0, DartFish, Alpharetta, GA, USA). Visual markers were placed on the left side of the body on the base of the fifth metatarsal, the lateral malleolus of the fibula, the joint line of the knee, the greater trochanter and the acromioclavicular joint. Ankle, knee, and hip joint angles were taken at the lowest position of each jump. Countermovement depth and vertical jump height were calculated as the vertical displacement of the marker on the greater trochanter from the reference of standing height.

All jumps were performed on a calibrated force plate (AccuPower, Athletic Republic, Fargo, ND, USA) with a sampling rate of 200 Hz. During each jump vertical RFD was calculated from the start of force development, defined as the point at which force exceeded 2% of the lowest value, to the following time points: 10, 10–30, 30–50, 50–100, 100–200, and 200–300 milliseconds. This approach allows for comparison of force development throughout different phases of the jump. Propulsion time was also calculated from the start of force development until toe off.

Statistical Analyses

The data are presented as mean $\pm$ SDs. All assumptions for linear statistics were met before further analysis. Paired *t*-tests were used to analyze the differences between jump types. A Bonferroni correction was used with these tests where appropriate. It was determined that a sample size of 25 was adequate to defend a 0.05 level of significance with a Cohen probability level of at least 0.8 for each dependent

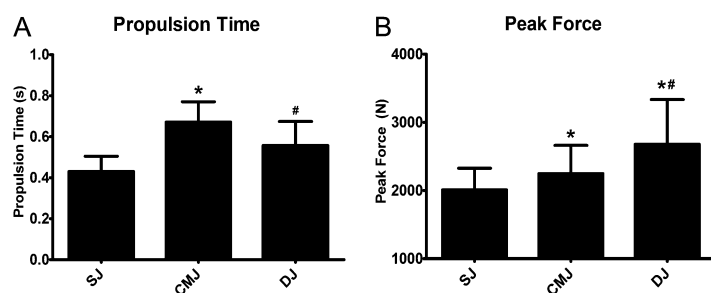


Figure 3. Peak vertical ground reaction force (A) and ground contact time (B) for static squat jumps (SJ), countermovement jumps (CMJ) and drop jumps off of a 30 cm box (DJ). *Indicates a significant ($p \leq 0.05$) difference from SJ. #Indicates a significant ($p \leq 0.05$) difference from CMJ.

variable (nQuery AdvisorO software; Statistical Solutions, Saugus, MA, USA) and use in regression analyses for stable beta coefficients. R -squared values have been reported to document the percentage of variance explained in the dependent by the independent variable, and the beta coefficient has been reported to show strength and direction of the relationship. Simple and stepwise regressions were used to determine the relationships between and among dependent variables. For this investigation, significance was defined as $p \leq 0.05$.

RESULTS

Difference between Jump Types

Jump height for DJ (57.0 ± 7.9 cm) and CMJ (56.6 ± 8.1 cm) was significantly greater than for SJ (54.17 ± 7.01 cm). However, there was no significant difference between DJ and CMJ. Peak vertical ground reaction force for DJ ($2,678 \pm 657$ N) was significantly greater than for CMJ ($2,250 \pm 413$ N) and

SJ ($2,009 \pm 319$ N), whereas peak force for CMJ was significantly greater than for SJ (Figure 3A). Propulsion time for CMJ (671 ± 99 milliseconds) was significantly greater than that for both SJ (430 ± 75 milliseconds) and DJ (557 ± 117 milliseconds) with no significant differences between SJ and DJ (Figure 3B). Although peak force was greater in CMJ than in SJ, RFD was only significantly greater in CMJ than in SJ at the later time points:

100–200 and 200–300 milliseconds time points (Figure 4). The DJ RFD was significantly greater than both CMJ and SJ at the earliest time points (0–10, 10–30, and 30–50 milliseconds) but was then significantly less than both jump types during the 50- to 100-millisecond time frame (Figure 5). During 100–200 milliseconds, there was no significant difference between DJ and either SJ or CMJ-RFD, while during 200- to 300-millisecond DJ, RFD was significantly less than CMJ but not different from SJ.

Prediction of Force Expression

No anatomical variables could predict propulsion time for any of the three jump types. See Table 2 for a summary of anatomical predictors of RFD during SJ, CMJ, and DJ. For SJs, AT length was an inverse predictor of RFD between 200 and 300 milliseconds ($r^2 = 0.251$, $\beta = -0.501$, $p = 0.013$). During CMJs, early force development could be predicted

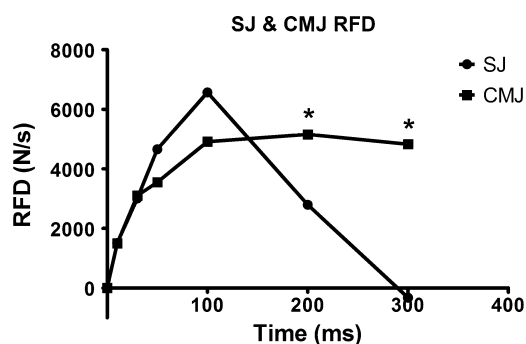


Figure 4. Rate of force development (RFD) from the start of vertical force development for static squat jumps (SJ) and countermovement jumps (CMJ). *Indicates a significant ($p \leq 0.05$) difference between variables.

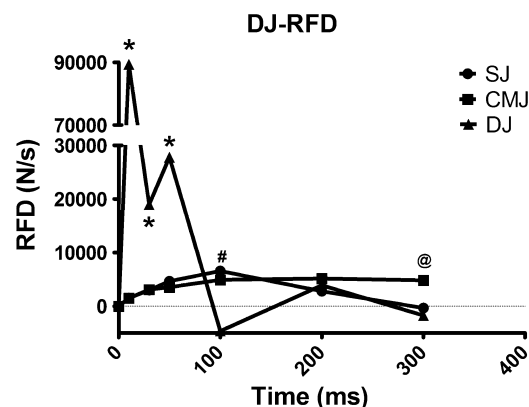


Figure 5. Rate of force development (RFD) of drop jumps (DJ) compared to that of static squat jumps (SJ) and countermovement jumps (CMJ). *Indicates DJ RFD is significantly ($p \leq 0.05$) greater than SJ and CMJ, #Indicates DJ RFD is significantly ($p \leq 0.05$) less than SJ and CMJ. @Indicates DJ RFD is significantly ($p \leq 0.05$) less than CMJ.

TABLE 2. Results from simple and stepwise regression analysis for 3 jumps with different prestretch loads from muscle and tendon structure.*

Jump type	Time (ms)	Step	Variable	β	Variance explained (%)	p
SJ	200–300	1	AT length	−0.501	25	0.013
CMJ	0–10	1	GAS-FL	0.461	21	0.020
		1	AT length	−0.438	19	0.029
DJ	10–30	1	GAS-FL	0.476	22	0.019
	0–10	1	GAS-FL	−0.485	24	0.014
	0–10	1	VL-PEN	0.435	19	0.030
	0–10	1	GAS-PEN	0.434	18	0.030
	0–10	1	GAS-FL	−0.462	24	0.010
		2	VL-PEN	0.410	40	0.021
	30–50	1	GAS thickness	0.440	19	0.028

*CMJ = countermovement jump; SJ = squat jump; DJ = drop jump; AT = Achilles tendon; GAS = gastrocnemius; FL = fascicle length; PEN = pennation angle.

between 0 and 10 milliseconds by either GAS-FL ($r^2 = 0.213$, $\beta = 0.461$, $p = 0.020$) or AT length ($r^2 = 0.191$, $\beta = -0.438$, $p = 0.029$), but there was no additive affect between the 2 variables. Between 10 and 30 milliseconds GAS-FL was a significant predictor of CMJ-RFD ($r^2 = 0.218$, $\beta = 0.476$, $p = 0.019$). During DJ, initial force development (0–10 milliseconds) could be significantly predicted by GAS-FL ($r^2 = 0.185$, $\beta = -0.434$, $p = 0.030$), VL-PEN ($r^2 = 0.189$, $\beta = 0.435$, $p = 0.030$), or GAS-PEN ($r^2 = 0.188$, $\beta = 0.434$, $p = 0.030$). This relationship became stronger by combining GAS-FL and VL-PEN ($r^2 = 0.403$, $\beta = -0.462$, 0.410 , $p = 0.021$). Lastly, between 30- to 50-milliseconds GAS cross-sectional thickness was a significant predictor of DJ RFD ($r^2 = 0.194$, $\beta = 0.440$, $p = 0.028$).

DISCUSSION

There are 3 major findings of this study, which provide important information regarding the influence of tendon, muscle FL, and PEN during SSC movements of varying intensities. (a) The AT length was an inverse predictor of RFD during initial force development during CMJ (10 milliseconds) and late RFD during SJ (200–300 milliseconds). (b) The GAS-FL had an intensity-dependent relationship with RFD during vertical jumping. (c) Subjects with greater VL and GAS-PEN were able to develop higher ground reaction forces during DJ landing.

Before discussion of the results, it is important to note that the strength of the predictions calculated in this study need to be taken in context. Muscle and tendon structures of the knee extensors and plantar flexors were unable to explain the majority of variance observed. The muscle and tendon structures of the plantar flexors alone were able to predict between 18 and 25% of the variance observed in RFD. In contrast, when supplementing plantar flexors with knee

extensor structures, the strength of prediction rose to 40%. Although these values may seem small at first, it should be noted that during vertical jumping, the estimated contributions of the plantar flexors and the knee extensors are 22 and 56%, respectively (25).

The first major finding of this study is that subjects with relatively long ATs had lower initial (0–10 milliseconds) RFD during CMJs. This was predicted based on the modeling study by Bobbert (11), which demonstrates that longer tendons would have greater elasticity than shorter tendons and thus have more time in which to reach maximal muscle tension. However, there was no observed increase in propulsion time or force production across any other time points. Furthermore, tendon length was not a significant predictor of RFD for either SJ or DJ. This may suggest there is an intensity-dependent effect of AT length on early force production. It is logical to infer that if spontaneous fascicle elongation of the GAS occurs (20,31) during early DJ, the AT would become unloaded and rapidly shorten, thus neglecting the influence of tendon compliance during this time point. In contrast, during SJ, it is possible that the initial force developed by muscles was dispersed through tendon displacement because of the difference in the mechanical properties of muscles and tendons. The delay in time between muscle activation and start of force development is called electromechanical delay. Previous research has shown that muscles with compliant tendons have longer electromechanical delay compared to stiffer tendons (18). This information combined with the results from this study would suggest that it is this increase in electromechanical delay that is important for increased jump performance while the actual propulsion time and RFD remain unchanged. Unfortunately, a limitation of this study is that although previous research has shown a strong relationship

between these variables, EMG and stiffness were not measured during this study. More research is needed to investigate if muscle activation time is influenced by tendon length and compliance.

Achilles tendon length was an inverse predictor of late (200–300 milliseconds) RFD during SJs, demonstrating that subjects with long tendons had a greater decline in RFD late in their propulsion. This finding supports the comments made above by inferring that, if early RFD was decreased because of increased elastic compliance, a delay in the force development would be expected in the force-time curve resulting in longer movement duration. However, because SJ propulsion time was not found to be related to tendon length such a shift was not observed in the current study. One possible explanation for why force production could no longer be maintained would be if greater work was done earlier in the jump because of increased early muscle activation and this interfered with the ability to maintain force production late in the movement. However, such a statement cannot be supported by this study because muscle activation was not measured.

The second major finding of this study was that there was an intensity-dependent relationship between GAS-FL and early force development with GAS-FL serving as a positive predictor of RFD during CMJ (0–10 and 10–30 milliseconds) but an inverse predictor during DJ (0–10 milliseconds). Previous research has stated that longer fascicles result in more contractile elements in series which increases the potential contractile velocity and the ability to produce force at high velocities while increasing the parallel elastic component of the muscle itself (5). Blazevich et al. (7) came to the conclusion that the increased parallel elastic component of long fascicles will lead to a decrease in early RFD (0–30 milliseconds) in isometric knee extensions. However, this study observed that greater GAS-FL was a significant predictor of CMJ-RFD between 0–10 and 10–30 milliseconds. So why does FL play a much different role in RFD during the CMJ than in isometric contractions? One possible explanation would be an increased contractile velocity of the muscle being beneficial because of the biarticular nature of the GAS. During the lowering phase of a CMJ, the muscle-tendon unit of the GAS undergoes simultaneous lengthening at the ankle and shortening at the knee. It is possible that despite the muscle itself acting in a concerted (or isometric) fashion during the CMJ, the contractile velocity of the muscle is important because it can allow for quicker buildup of tension on the tendons and overcome the rapid shortening of the muscle-tendon unit resulting from knee flexion. A second possible explanation of this relationship would stem from neurological influences resulting from a short latency stretch reflex (SLSR). It is possible that the higher elastic compliance of long fascicles may result in significantly greater initial fascicle strain during initial eccentric contractions. However, the active strain of

intrafusal fibers can increase type 1a and type 2 afferent signals and result in an increased SLSR, which has previously been shown to occur 10–12 milliseconds after the start of muscle activation (24).

The third major finding of this study is that subjects with highly pennated VL and GAS and short GAS fascicles have greater early RFD during DJs, which suggests that they are better equipped to handle high eccentric loads. It has been previously stated that muscle PEN is beneficial for force production during slow strength movements as PEN increases the physiological cross-sectional area of a muscle (5,26). However, because of the indirect line of pull of the fibers, increased muscle PEN has been shown to decrease force per cross-sectional area (21). It has been previously hypothesized that because of this indirect line of pull of pennated muscles, the muscle will have an increased ability to resist external forces (15). This occurs because direct tendon forces acting on the muscle are dissipated by a factor of cosine of the angle of PEN. Muscle PEN also increases the viscosity of the muscle by increasing shearing forces and distributing strain forces to a greater surface area along the aponeurosis (15).

During DJs, GAS fascicle behavior is dependent on the height over which the drop occurs (19,20). Trained individuals respond positively in jump performance and power production to increased drop heights until a threshold height is reached and then performance plateaus or decreases (13,29). Similarly, GAS fascicle behavior changes because supramaximal jumps have been associated with an uncontrollable fascicle lengthening. The current results give further support to those recently published (15), which reported subjects with long fascicles to have a significantly decreased jump performance during DJs when compared to SJ and CMJ. It was concluded that longer GAS fascicles have a decreased capacity to handle high eccentric loads, which decreases the height at which uncontrolled GAS lengthening occurs (14). This study supports this conclusion finding that subjects with long GAS-FL had significantly lower initial RFD during DJs off a 30-cm box.

PRACTICAL APPLICATIONS

Interpretation of previous work showing long compliant tendons was better suited for jumping should take into account that during the quick jump situations often observed in sport that tendon length may actually diminish rather than enhance performance, and thus decrease the importance of AT-length for talent identification. Optimal muscle architecture appears to be dependent on both the eccentric load and the phase of jump. Although both strength and plyometric training has been shown to increase FL, only heavy resistance training has been shown to increase PEN. Thus, when a high eccentric load or multiple jumps are required for sport, heavy strength training should be used to allow for early force production during jumping.

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